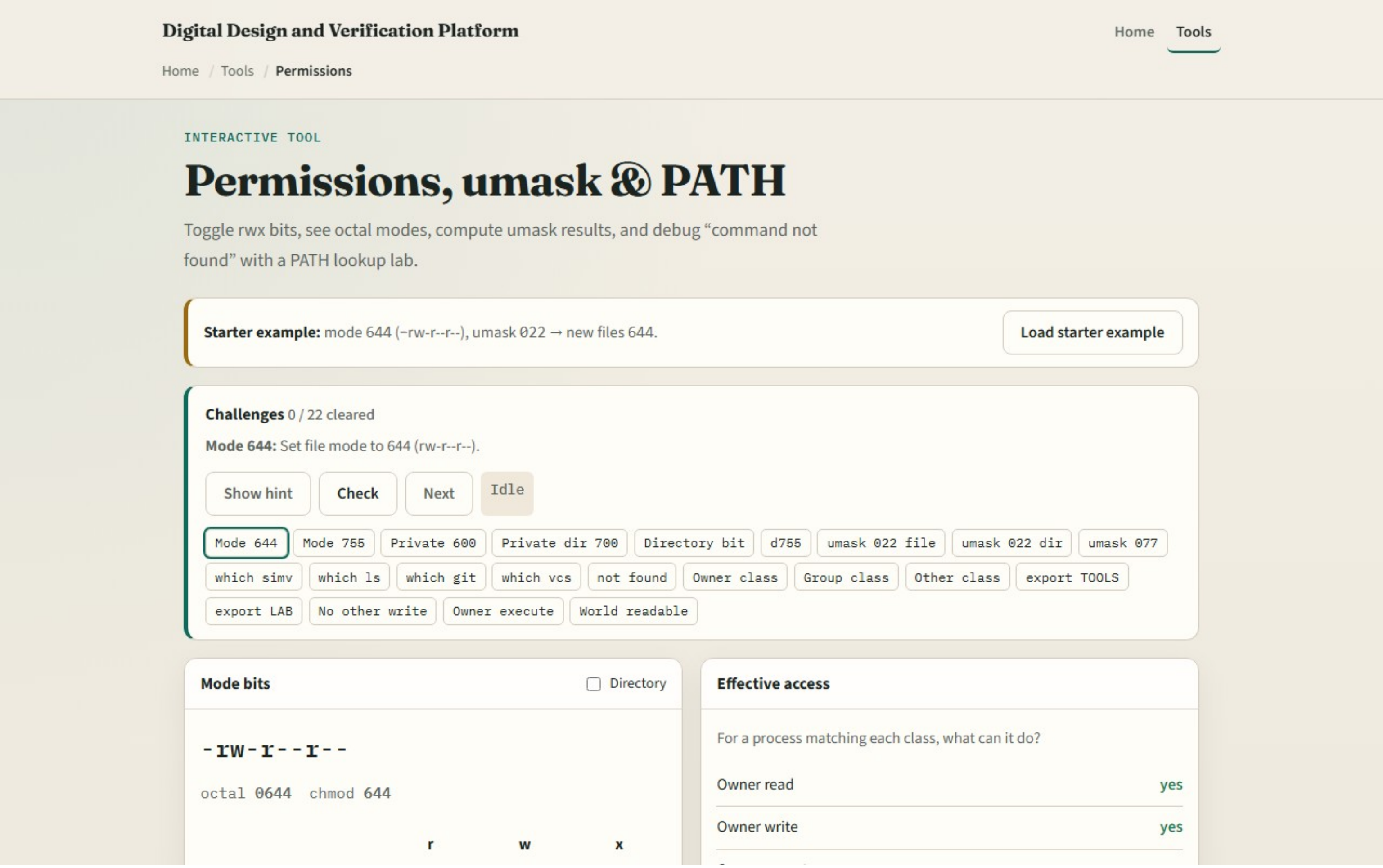
Permissions, umask, PATH

Who can read a file, who can run a script

Mode bits, umask, and PATH

- In a long listing, the nine permission letters are user, group
- Chmod changes those bits; a script needs execute before you can run it with dot-slash
- Umask masks bits off new files so you do not accidentally create world-writable junk
- PATH is a colon-separated search list

Browser lab



Permissions lab

Real shell practice

```
wsl — module07-permissions/examples/permissions

# Track A — real Unix
# real shell session (Track A)

$ mkdir -p /tmp/m07_perms && cp -f hello.sh config.txt /tmp/m07_perms/ && cd /tmp/m07_perms && chmod 644 hello.sh
config.txt && ls -l
total 8
-rw-r--r-- 1 yongfu yongfu 145 Jul 20 03:03 config.txt
-rw-r--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ chmod u+x hello.sh

$ ls -l hello.sh
-rwxr--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ ./hello.sh
Hello from hello.sh — permissions example works!

$ chmod 600 config.txt

$ ls -l
total 8
-rw------- 1 yongfu yongfu 145 Jul 20 03:03 config.txt
-rwxr--r-- 1 yongfu yongfu 150 Jul 20 03:03 hello.sh

$ umask
0022

$ umask -S
u=rwx,g=rx,o=rx
```

Real shell — permissions and umask

Real shell practice — try these

```
# ls -l — long listing; read the nine permission letters
ls -l

# chmod u+x hello.sh — add execute for your user (symbolic mode)
chmod u+x hello.sh

# ls -l hello.sh — confirm the user execute bit appears
ls -l hello.sh

# ./hello.sh — run the script in this directory (dot is usually not on
PATH)
./hello.sh

# chmod 600 config.txt — owner read/write only (numeric mode)
chmod 600 config.txt
```

Pitfalls to watch

- Avoid chmod seven-seven-seven unless you truly understand the risk
- Prefer user-plus-execute for scripts you own
- Remember: without dot-slash
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, practice chmod, umask
- When you are ready, take the short quiz, then continue to dotfiles and config homes